

A universal group of type F_∞

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Abstract

We construct a single group of type F_∞ containing every finitely presented group. Its finitely generated subgroups, up to isomorphism, are exactly the finitely generated recursively presented groups. This answers the F_∞ form of the higher-dimensional Higman embedding question.

1 Introduction

A group has type F_n if it admits a classifying CW complex with finite n -skeleton, and type F_∞ if it admits one classifying complex with finitely many cells in every dimension. Here a classifying complex is a connected CW complex with the given fundamental group and contractible universal cover; it may have infinitely many dimensions. Type F_2 is equivalent to finite presentability.

Theorem 1.1. *There exists a group H of type F_∞ such that every finitely presented group admits an injective homomorphism into H .*

The common target gives an exact description of its finitely generated subgroups. A finitely generated group is recursively presented if it admits a presentation on finitely many generators with a recursively enumerable set of relators.

Corollary 1.2. *A finitely generated group embeds in H if and only if it is recursively presented.*

Proof. If G is finitely generated and recursively presented, Higman's embedding theorem places it in a finitely presented group [6], which embeds in H by Theorem 1.1. Conversely, H is finitely presented because it has type F_∞ . For a finitely generated subgroup $G \leq H$, represent its generators by words in a finite presentation of H . Enumerate all words in the chosen generators, and dovetail finite derivations from the ambient relators proving that their substituted words are trivial. Recording the successful words enumerates exactly the defining relators of G . $\square$

This answers the F_∞ form of the higher-dimensional Higman embedding question, stated for finitely generated recursively presented groups in [4, Question 1.3]. The theorem imposes no word-problem hypothesis on the input.

Context and construction

For a commutative unital coefficient ring Λ , a group G has type $FP_n(\Lambda)$ if the trivial ΛG -module Λ has a projective resolution with finitely generated terms through degree n ; write FP_n for $\Lambda = \mathbb{Z}$. For $n = \infty$, finite generation is required in every degree. Leary proved that every countable group embeds in a group of type FP_2 [8]. Fournier-Facio and Zaremsky show that, for each $n \in \mathbb{N} \cup \{\infty\}$, embeddings of all finitely generated recursively presented groups into recursively presented groups

of type FP_n would imply embeddings of all finitely presented groups into groups of type F_n [4, Theorem A]. They also prove that the rope-trick groups of their Definition 1.5 fail type $FP_3(\mathbb{Q})$ for infinite inputs with nontrivial free-presentation kernel [4, Definition 1.5 and Theorem B]. This obstruction is specific to those rope-trick groups.

The proof of their conditional theorem uses a universal finitely presented group, an embedding through an intermediate overgroup and back, and an ascending HNN extension. They relate this strategy to earlier work of Baumslag–Dyer–Miller and Fournier–Facio–Löh–Moraschini [4, Section 2].

For the universal finitely presented group U of Lemma 2.2, Proposition 4.1 gives a finite presentation of a group $\widehat{W}$ equipped with two finite families of homomorphisms from U , sharing a distinguished map $\iota : U \rightarrow \widehat{W}$. Faithful models in tensor product algebras verify that ι is injective. Finite presentability of $\widehat{W}$ then lets universality supply an embedding $\kappa : \widehat{W} \rightarrow U$:

$$\begin{array}{ccc} U & \xrightarrow{\iota} & \widehat{W} \xrightarrow{\kappa} U \\ & \searrow f=\kappa\iota & \nearrow \end{array}$$

Postcomposing the two families by κ gives homomorphisms from U to itself, with the injective endomorphism f as their common distinguished map (Corollary 4.2). The group U embeds in the associated ascending HNN extension

$$H = T(U, f) = \langle U, t \mid t^{-1}ut = f(u) \ (u \in U) \rangle,$$

which is finitely presented by Lemma 5.2. Theorem 5.1 proves that H has type F_∞ .

The finite idempotent systems, the group enlargement, and the ascending-torus criterion are shared with the companion [9, Sections 2.1, 2.3–2.5, and 3.1]. We give local proofs of the algebraic identities and geometric criterion.

The geometric step begins with a uniform filling statement for ordinary integral chains. In the homogeneous chain model, restricting relative vertex differences to a finite subset of U leaves only finitely many simplex types up to common left translation in each degree. For each positive degree and finite input set of relative differences, Proposition 6.2 chooses one exponent and one finite output set. That iterate of f sends every integral cycle using the input differences to the boundary of a chain using the output differences. The filling lengths and coefficients may vary with the cycle.

The finite diagrams supply these fillings by induction. For each $\ell \in \{2, 3\}$, the diagram is indexed by idempotents over $\mathbb{F}_\ell$. Orthogonal sums correspond to pointwise products of homomorphisms with elementwise commuting images. Lemma 3.1 says that an additive assignment has its value at the identity annihilated by ℓ . In degrees $n \geq 2$, the chain diagonal includes mixed terms from lower degrees, so the corresponding cycle classes are not automatically additive. Using lower-degree fillings, the induction constructs, below degree n , a controlled chain homotopy from the map induced by an iterate of f to the constant map. Lemma 6.1 uses it to remove the mixed terms. For each input cycle, both diagrams then give additive assignments into the homology of the same bounded chain complex. Their common value at the identity idempotent is represented by the image of the input cycle under an iterate of f . It is annihilated by both 2 and 3, and hence vanishes.

For an arbitrary set $\mathcal{I}$, put $M = (\mathbb{Z}H)^\mathcal{I}$. A cycle in the tensor complex computing $H_i(U, M)$, $i > 0$, decomposes in each coordinate into finitely many ordinary cycles in left U -cosets, with one finite set of simplex types common to all coordinates. The uniform filling statement supplies one exponent and output set for these cycles, so their fillings reassemble into one tensor chain. In the ascending HNN sequence of Section 7, the map F combines f on chains with multiplication by the stable letter on coefficients. The fillings make F locally nilpotent on $H_i(U, M)$ for $i > 0$: every class is killed by some power of F . Thus $1 - F$ is invertible there, and exactness gives $H_n(H, M) = 0$ for

every $n \geq 2$. Since H is finitely presented, Lemma 7.1 turns this vanishing into geometric finiteness. At each stage, the cellular cycle module is finitely generated; via Hurewicz, those generators have sphere representatives. Attaching cells along these spheres stage by stage produces one classifying complex with finitely many cells in every dimension.

Conventions

All groups are discrete. Group homology has integral coefficients unless stated otherwise, and coefficient modules are right modules. Products indexed by arbitrary sets always mean direct products, not direct sums.

2 A universal finitely presented group

We will place a finite collection of homomorphisms inside one finitely presented overgroup and then embed that overgroup back into the original group. For this purpose we need a finitely presented group containing every finitely presented group. We recall its construction from Higman's embedding theorem [6]: every finitely generated group with a recursively enumerable presentation embeds in a finitely presented group. No decidability assumption on the word problem occurs in this theorem.

The base-group embedding and finite-generation construction below follow the classical embedding methods of Higman, B. H. Neumann and Hanna Neumann [7, Theorems I, I', and IV]; we include the arguments in the form needed here. We first record an elementary form of base-group injectivity for HNN extensions. Here the countable case is sufficient.

Lemma 2.1. *Let D be a countable group, let D_1, D_2 be subgroups of D , and let $\theta : D_1 \rightarrow D_2$ be an isomorphism. The natural homomorphism from D to*

$$\langle D, t \mid t^{-1}dt = \theta(d) \quad (d \in D_1) \rangle$$

is injective. Here the presentation includes all relations of D .

Proof. Let D act faithfully on $\Omega = D \times \mathbb{N}$ by left multiplication in the first coordinate. The restrictions to D_1 and D_2 are free actions, each with countably infinitely many orbits. Match their orbit sets bijectively. For each matched pair choose a base point x in the D_2 -orbit and a base point y in the D_1 -orbit, and define

$$T(\theta(d)x) = dy \quad (d \in D_1).$$

Freeness and the orbit matching make T a permutation of Ω . For every $d \in D_1$ it satisfies $T\theta(d) = dT$, and hence $T^{-1}dT = \theta(d)$, as permutations of Ω . Sending t to T therefore defines an action of the displayed group whose restriction to D is faithful. The natural map from D must be injective. $\square$

Lemma 2.2. *There exists a finitely presented group U into which every finitely presented group embeds.*

Proof. Effectively list all finite presentations, renaming their generators so that the generating alphabets are disjoint. Let C be the group given by the union of these presentations. Its generators can be effectively numbered $c_0, c_1, \dots$, and its relators can be effectively enumerated. Each group on the list embeds in C : the homomorphism induced by its generators has a left inverse obtained by killing all generators of the other presentations.

It remains to place C inside a finitely generated group with recursively enumerable relators. Form

$$D = C * \langle a, p \rangle, \quad v_i = a^i p a^{-i} \quad (i \geq 0),$$

where $\langle a, p \rangle$ is free of rank two. The elements v_i freely generate their subgroup: in any product of nonzero powers of the v_i with consecutive indices distinct, consecutive powers of p are separated by nonzero powers of a , so free reduction cannot give the identity. The elements $c_i v_i$ also freely generate their subgroup, because the projection $D \rightarrow \langle a, p \rangle$ killing C sends them to the v_i . Consequently $v_i \mapsto c_i v_i$ extends to an isomorphism between these two free subgroups of D .

By Lemma 2.1, D , and hence C , embeds in

$$V = \langle D, q \mid q^{-1} v_i q = c_i v_i \quad (i \geq 0) \rangle.$$

The identities

$$c_i = q^{-1} v_i q v_i^{-1}$$

show that V is generated by a, p, q . More precisely, it has a presentation on these three generators obtained by substituting the displayed word for every c_i in the relators of C . The substitution turns each HNN relation into an identity. Thus it and the natural map into V are mutually inverse homomorphisms, so this is indeed a presentation of V . Its relators are recursively enumerable. Higman's embedding theorem now embeds V in a finitely presented group U . Every finitely presented group embeds in C , then in V , and finally in U . $\square$

Below we use only the existence of the subgroup embeddings supplied by Lemma 2.2.

3 Two finite systems of idempotents

The finite diagrams used in our construction are indexed by idempotent operators over $\mathbb{F}_2$ and $\mathbb{F}_3$. Their essential property is elementary: an additive assignment to either diagram forces the value at the identity to be annihilated by the corresponding prime. These diagrams and their additivity argument are shared with the companion [9, Section 2.1, Lemma 2.1]. We give the complete argument before realizing them by homomorphisms of groups.

For $\ell \in \{2, 3\}$, let D_ℓ be the unital $\mathbb{F}_\ell$ -algebra of operators on the free vector space with basis $\{0, 1\}^\mathbb{N}$ generated by l_0, l_1, r_0, r_1 . Here l_i prefixes the symbol i , and r_i deletes an initial i , sending basis vectors with the other initial symbol to zero. Writing I for the identity operator, we have

$$r_i l_j = \delta_{ij} I, \quad l_0 r_0 + l_1 r_1 = I. \quad (3.1)$$

For a finite binary word v , let l_v prefix v , let r_v delete v when it occurs as a prefix and give zero otherwise, and put $E_v = l_v r_v$. Thus E_v projects onto the span of the sequences beginning with v .

If (v_1, v_2, v_3) is a prefix-free list, then

$$(l_{v_i} r_{v_j})(l_{v_k} r_{v_m}) = \delta_{jk} l_{v_i} r_{v_m}.$$

These operators are nonzero matrix units, so their span is a copy of $M_3(\mathbb{F}_\ell)$. We use three such algebras, connected by shared diagonal projections. Write

$$A = E_{00}, \quad A_0 = E_{000}, \quad A_1 = E_{001}, \quad B' = E_{01}, \quad C' = E_{1}.$$

Their defining lists, diagonal projections, and identity operators are

prefix list	diagonal projections	identity
(00, 01, 1)	A, B', C'	I
(000, 01, 1)	A_0, B', C'	$A_0 + B' + C'$
(001, 01, 1)	A_1, B', C'	$A_1 + B' + C'$

(3.2)

The last two identities are not I , while $A = A_0 + A_1$ and $A_0A_1 = A_1A_0 = 0$. Let $\mathcal{P}_\ell$ be the union, as a set of actual operators in D_ℓ , of all idempotents in these three matrix algebras. It is finite and contains I . An operator belonging to two of the algebras has just one index in $\mathcal{P}_\ell$.

Two operators E, P are called *orthogonal* if $EP = PE = 0$. The relations below include all orthogonal pairs whose sum belongs to $\mathcal{P}_\ell$, even when the pair does not belong to a single one of the three matrix algebras. In particular, they include $A = A_0 + A_1$, whose summands come from the two smaller support algebras.

Lemma 3.1. *Let $\ell \in \{2, 3\}$, let T be an abelian group, and let $(z_E)_{E \in \mathcal{P}_\ell}$ be elements of T . Suppose that*

$$z_{E+P} = z_E + z_P \quad (3.3)$$

whenever $E, P, E + P \in \mathcal{P}_\ell$ and $EP = PE = 0$. Then $\ell x_I = 0$.

Proof. We first compare the values of rank-one idempotents within any one of the three matrix algebras. For a column vector $u \in \mathbb{F}_\ell^3$ and a row covector λ with $\lambda u = 1$, write

$$x(u, \lambda) = z_{u\lambda}.$$

Fix a nonzero vector $v \in \ker \lambda$. We claim that the difference

$$x(u + v, \lambda) - x(u, \lambda) \quad (3.4)$$

does not depend on the vector u subject to $\lambda u = 1$.

Take another such vector u' . Suppose first that $u' - u$ and v are linearly independent. They form a basis of $\ker \lambda$, and hence $u, u' - u, v$ form a basis of $\mathbb{F}_\ell^3$. There is consequently a covector μ satisfying

$$\mu u = \mu u' = 0, \quad \mu v = 1.$$

The identity

$$u\lambda + v\mu = (u + v)\lambda + v(\mu - \lambda)$$

expresses one rank-two idempotent as two orthogonal sums of rank-one idempotents. Indeed, $\lambda v = \mu u = 0$ on the left, while $\lambda v = (\mu - \lambda)(u + v) = 0$ on the right; all four rank-one factors have their defining covector equal to one on their defining vector. Applying (3.3) gives

$$x(u + v, \lambda) - x(u, \lambda) = x(v, \mu) - x(v, \mu - \lambda). \quad (3.5)$$

The same covector μ gives the same right-hand side for u' , proving the claim in this case.

If $u' - u$ lies in the span of v , choose $w \in \ker \lambda \setminus \langle v \rangle$ and put $u'' = u + w$. Both $u'' - u$ and $u'' - u'$ are independent of v . Comparing first u with u'' and then u' with u'' proves the claim in general. The kernel of λ has dimension two, so this argument also works over $\mathbb{F}_2$; no choice of a third nonzero scalar is being used.

Apply the claim to the successive vectors $u, u + v, \dots, u + (\ell - 1)v$. The ℓ successive differences are equal, and they sum to zero because $\ell v = 0$. Therefore

$$\ell x(u + v, \lambda) = \ell x(u, \lambda). \quad (3.6)$$

The same conclusion is immediate for $v = 0$, so $\ell x(u, \lambda)$ is independent of u whenever λ is fixed and $\lambda u = 1$. For the covector comparison, work within this same matrix algebra and define a new assignment $z'_E = z_{E^\top}$. Transposition preserves idempotence and orthogonal sums, so z' obeys the same additivity relations. Apply the invariance just proved to $x'(v, \mu) = z'_{v\mu} = z_{\mu^\top v^\top}$. Fixing μ and

varying v now shows that $\ell x(u, \lambda)$ is independent of λ whenever u is fixed and $\lambda u = 1$. This does not assume that $z_E = z_{E^\top}$.

In particular the three diagonal matrix units have equal ℓ -multiples. To see the comparison explicitly, for distinct basis vectors e_i, e_j take $\lambda = e_i^* + e_j^*$. First change the covector at e_i from e_i^* to λ , then change the vector from e_i to e_j , and finally change the covector to e_j^* . Each change preserves the ℓ -multiple, giving

$$\ell z_{e_i e_i^*} = \ell z_{e_j e_j^*}.$$

It remains to use the shared projections displayed above. The comparison of diagonal units in each algebra gives

$$\ell z_A = \ell z_{B'} = \ell z_{C'} = \ell z_{A_0} = \ell z_{A_1}.$$

Call this common value $q \in T$. The projection identity $A = A_0 + A_1$ is an orthogonal sum across the two smaller support algebras. Its three operators belong to $\mathcal{P}_\ell$, so (3.3) gives $q = q + q$. Thus $q = 0$. Finally, $A + B'$ belongs to the first matrix algebra, and $I = (A + B') + C'$. Applying additivity twice yields

$$\ell z_I = \ell z_A + \ell z_{B'} + \ell z_{C'} = 0,$$

as required. $\square$

4 Faithful finite diagram envelopes

The finite idempotent sets $\mathcal{P}_2$ and $\mathcal{P}_3$ specify only finitely many commutation and multiplication identities. We now realize all these identities in a finitely presented overgroup, without collapsing the common full-identity copy of the input group. A faithful model supplies the injectivity; its own presentation need not be finite.

Proposition 4.1. *For every finitely presented group U there are a finitely presented group $\widehat{W}$ and homomorphisms*

$$k_{\ell, E} : U \longrightarrow \widehat{W} \quad (\ell \in \{2, 3\}, E \in \mathcal{P}_\ell)$$

such that $k_{2, I} = k_{3, I}$ is injective. For each ℓ , whenever $E, E', E + E' \in \mathcal{P}_\ell$ and $EE' = E'E = 0$, the subgroups $k_{\ell, E}(U)$ and $k_{\ell, E'}(U)$ commute elementwise, and

$$k_{\ell, E+E'}(u) = k_{\ell, E}(u)k_{\ell, E'}(u) \quad (u \in U).$$

Proof. Fix a finite presentation $U = \langle X \mid \mathcal{R} \rangle$. For every pair (ℓ, E) take a copy of this presentation, with generators $x_{\ell, E}$ for $x \in X$. Define $\widehat{W}$ by these copy presentations and the following additional relations:

$$x_{2, I} = x_{3, I} \quad (x \in X), \quad (4.1)$$

$$[x_{\ell, E}, y_{\ell, E'}] = 1 \quad (x, y \in X), \quad (4.2)$$

$$x_{\ell, E+E'} = x_{\ell, E} x_{\ell, E'} \quad (x \in X). \quad (4.3)$$

The last two families run over both values of ℓ and every orthogonal pair E, E' whose members and sum lie in $\mathcal{P}_\ell$. All three lists are finite. The indices are actual idempotents in D_ℓ , not separately tagged copies from its three matrix subalgebras. Thus these relations include the orthogonal sums between different subalgebras as well as those within one subalgebra. In particular, the zero idempotent is included; its sum relation with itself gives $x_{\ell, 0} = x_{\ell, 0}^2$, forcing $x_{\ell, 0} = 1$.

The copy presentations give homomorphisms $k_{\ell, E} : U \rightarrow \widehat{W}$. Equation (4.2) implies that their corresponding image subgroups commute elementwise. It follows that $u \mapsto k_{\ell, E}(u)k_{\ell, E'}(u)$ is a

homomorphism. By Equation (4.3), it agrees with $k_{\ell, E+E'}$ on X , and therefore on all of U . Similarly, Equation (4.1) gives $k_{2,I} = k_{3,I}$.

To prove that this common map is injective, we construct a faithful model. The same group-level enlargement is used in the companion [9, Section 3.1]; we verify all of its needed algebraic properties here. For any group V and $\ell \in \{2, 3\}$, work in the unital algebra

$$A_\ell(V) = D_\ell \otimes_{\mathbb{F}_\ell} \mathbb{F}_\ell[V],$$

where $\mathbb{F}_\ell[V]$ is the group algebra with basis indexed by V . For $E \in \mathcal{P}_\ell$ define

$$j_{\ell, E}(v) = (I - E) \otimes 1 + E \otimes v \quad (v \in V). \quad (4.4)$$

Since E and $I - E$ are orthogonal idempotents, multiplication gives

$$j_{\ell, E}(v)j_{\ell, E}(w) = j_{\ell, E}(vw), \quad j_{\ell, E}(1) = 1.$$

Thus these elements are units, with inverses $j_{\ell, E}(v^{-1})$, and $j_{\ell, E}$ is a homomorphism.

For $E, E', E + E' \in \mathcal{P}_\ell$ with $EE' = E'E = 0$, rewrite Equation (4.4) as $1 + E \otimes (v - 1)$. Thus

$$(1 + E \otimes (v - 1))(1 + E' \otimes (w - 1)) = 1 + E \otimes (v - 1) + E' \otimes (w - 1),$$

because $EE' = 0$. Reversing the order gives the same answer since $E'E = 0$. Hence the images of $j_{\ell, E}$ and $j_{\ell, E'}$ commute elementwise, even if v and w do not commute. Setting $w = v$ gives

$$j_{\ell, E}(v)j_{\ell, E'}(v) = j_{\ell, E+E'}(v).$$

Moreover, $j_{\ell, I}(v) = I \otimes v$ is injective: the decomposition $A_\ell(V) = \bigoplus_{v \in V} D_\ell \otimes v$ distinguishes these elements, since $I \neq 0$.

Let $W_\ell(V)$ be the subgroup of $A_\ell(V)^\times$ generated by all images $j_{\ell, E}(V)$. First form $W_2(U)$, and then form $W_3(W_2(U))$. Write $j_E^{(2)}$ for the first-stage maps and $j_E^{(3)}$ for the second-stage maps. The group $W_3(W_2(U))$ receives the two families

$$\alpha_E = j_I^{(3)} \circ j_E^{(2)} \quad (E \in \mathcal{P}_2), \quad \beta_E = j_E^{(3)} \circ j_I^{(2)} \quad (E \in \mathcal{P}_3).$$

Postcomposition preserves the first family's identities, and precomposition preserves the second family's identities. Their full-identity maps coincide with the injection $j_I^{(3)} \circ j_I^{(2)}$. Notice that the second group algebra is formed from the abstract group $W_2(U)$; no map between algebras of different characteristics is involved.

These families satisfy every defining relation of $\widehat{W}$. They therefore induce a homomorphism $\Phi : \widehat{W} \rightarrow W_3(W_2(U))$ for which

$$\Phi \circ k_{2,I} = j_I^{(3)} \circ j_I^{(2)}.$$

The right-hand side is injective, so $k_{2,I}$ is injective as claimed. $\square$

We can now bring the entire finite diagram back into a universal group.

Corollary 4.2. *Let U be a finitely presented group containing an isomorphic copy of every finitely presented group. There are an injective endomorphism $f : U \rightarrow U$ and homomorphisms $h_{\ell, E} : U \rightarrow U$ for $\ell \in \{2, 3\}$ and $E \in \mathcal{P}_\ell$, with $h_{2,I} = h_{3,I} = f$, satisfying all the orthogonal commutation and sum identities of Proposition 4.1.*

Proof. Apply Proposition 4.1 to U . Since $\widehat{W}$ is finitely presented, universality supplies an embedding $\kappa : \widehat{W} \rightarrow U$. Set

$$h_{\ell, E} = \kappa \circ k_{\ell, E}, \quad f = \kappa \circ k_{2,I} = \kappa \circ k_{3,I}.$$

The map f is injective, and postcomposition by κ preserves every required identity and commutation relation. $\square$

5 The ascending-torus theorem

The finite diagrams constructed above will supply all the higher finiteness properties at once. We now state the theorem that makes this possible and give its proof, including the passage from homology to a classifying space with finite skeleta. This ascending-torus criterion and its homological construction are shared with the companion [9, Theorem 2.2 and Sections 2.4–2.5].

For a group U and an endomorphism $f : U \rightarrow U$, define

$$T(U, f) = \langle U, t \mid t^{-1}ut = f(u) \text{ for all } u \in U \rangle.$$

This presentation includes all the relations of U . When f is injective, it is the ascending HNN extension associated to f .

Theorem 5.1 (Ascending-torus theorem). *Let U be a group and $f : U \rightarrow U$ an injective endomorphism. Suppose that the following conditions hold.*

- (i) *There are a finitely presented group P and homomorphisms $a : U \rightarrow P$ and $b : P \rightarrow U$ such that $f = ba$.*
- (ii) *For each $\ell \in \{2, 3\}$ there are homomorphisms $h_{\ell, E} : U \rightarrow U$, indexed by the finite set $E \in \mathcal{P}_\ell$ defined above, with $h_{\ell, I} = f$. Whenever $E, E', E + E' \in \mathcal{P}_\ell$ and $EE' = E'E = 0$, the subgroups $h_{\ell, E}(U)$ and $h_{\ell, E'}(U)$ commute elementwise, and*

$$h_{\ell, E+E'}(u) = h_{\ell, E}(u)h_{\ell, E'}(u) \quad (u \in U).$$

Then U embeds naturally in $T(U, f)$, and $T(U, f)$ is of type F_∞ .

The proof has three parts. First, factoring through a finitely presented group gives a finite presentation of the ascending torus. Second, the two idempotent diagrams give uniformly controlled fillings of ordinary integral cycles after applying an iterate of f . Finally, these fillings imply vanishing of homology with products of regular coefficient modules. A cellular argument then constructs one classifying space with finitely many cells in every dimension. No word-problem hypothesis is used in any of these steps.

Lemma 5.2. *Suppose that $f = ba$ for homomorphisms $a : U \rightarrow P$ and $b : P \rightarrow U$, where P is finitely presented. Then $T(U, f)$ is finitely presented. This assertion does not require f to be injective.*

Proof. Put $g = ab : P \rightarrow P$, and write s for the stable letter of $T(P, g)$. Define maps on the respective generating groups and stable letters by

$$\begin{aligned} \Phi : T(U, f) &\longrightarrow T(P, g), & u &\longmapsto a(u), & t &\longmapsto s, \\ \Psi : T(P, g) &\longrightarrow T(U, f), & p &\longmapsto tb(p)t^{-1}, & s &\longmapsto t. \end{aligned}$$

For the first map, $af = ga$ verifies the defining conjugation relations. For the second, the image of $s^{-1}ps$ is $b(p)$, whereas the image of $g(p)$ is

$$tb(g(p))t^{-1} = tf(b(p))t^{-1} = b(p).$$

Thus both are homomorphisms. Their composites fix the stable letters and satisfy

$$\Phi\Psi(u) = tf(u)t^{-1} = u, \quad \Phi\Psi(p) = sg(p)s^{-1} = p.$$

Consequently $T(U, f) \cong T(P, g)$. If $P = \langle y_1, \dots, y_r \mid R_1, \dots, R_m \rangle$, choose a word w_i representing $g(y_i)$ for each i . The finite presentation

$$\langle y_1, \dots, y_r, s \mid R_1, \dots, R_m, s^{-1}y_i s = w_i \ (1 \leq i \leq r) \rangle$$

presents $T(P, g)$. No injectivity of g is required. $\square$

The remaining task is therefore higher finiteness. The next section proves the uniform filling assertion; Section 7 uses it to complete the proof of Theorem 5.1.

6 Uniform fillings with finitely many simplex types

We now prove the chain-level part of Theorem 5.1. Its purpose is to obtain one iterate of f and one finite set of allowable simplex types that fill every cycle of a specified dimension and input bound. The companion [9, Lemma 2.5 and Proposition 2.6] uses the same homological device. It requires neither uniformly short fillings nor an algorithm for finding them.

6.1 The homogeneous complex and its bounds

Let $\mathcal{E}(U)$ be the simplicial set whose q -simplices are all tuples $(u_0, \dots, u_q)$ of elements of U . Faces delete entries, and degeneracies repeat entries. Write $C_*(U)$ for its normalized integral chain complex. Thus its basis in degree q consists of tuples with no two consecutive entries equal; a tuple with consecutive equal entries represents zero. The differential is zero in degree zero, and, for $q \geq 1$, it is

$$\partial(u_0, \dots, u_q) = \sum_{i=0}^q (-1)^i (u_0, \dots, \widehat{u_i}, \dots, u_q).$$

The augmentation $\epsilon : C_0(U) \rightarrow \mathbb{Z}$ sends every vertex to one. Prepending the identity vertex contracts the augmented complex as a complex of abelian groups. Simultaneous left translation gives a free U -action on the normalized basis, so the augmented complex is also a free resolution of the trivial left $\mathbb{Z}U$ -module. The contraction need not be U -equivariant.

A *bound* is a finite symmetric subset $A \subseteq U$ containing 1. Let $\mathcal{E}_A(U)$ be the simplicial subset of tuples with

$$u_i^{-1}u_j \in A \quad \text{for all } i, j,$$

and denote its normalized chains by $C_*^A(U)$. The condition is preserved by deleting or repeating vertices, so this is a chain subcomplex. A reduced zero-cycle means an element of the kernel of ϵ ; in positive degree, “reduced cycle” means an ordinary cycle.

In each fixed degree, $C_*^A(U)$ has only finitely many basis elements up to left translation. Indeed, translating the first vertex to 1 puts every remaining vertex in A . These orbits are the *simplex types* of bound A . Notice the distinction: a bound restricts each simplex’s relative vertices, not the locations of its vertices in U , the number of simplices in a chain, or its integer coefficients.

A homomorphism of groups sends a finite bound to a finite bound. We say that a chain operator on $C_*^A(U)$ is *controlled* through specified degrees if there is a finite bound containing every simplex in all of its output chains in those degrees. For finitely many operators and degrees, separate output bounds may be replaced by their union, with inverses and 1 adjoined. For chains in $\mathcal{E}(U \times U)$, we will also use separate finite bounds in the two coordinates.

6.2 A controlled additivity calculation

Let $c : C_*(U) \rightarrow C_*(U)$ denote the chain map of the constant vertex map $u \mapsto 1$. Thus c_0 sends every vertex to (1) , and $c_q = 0$ for $q > 0$. The next lemma is an ordinary chain calculation: its homotopies are not required to commute with the U -action.

Lemma 6.1. *Let $n \geq 1$, let A be a bound, and let $f : U \rightarrow U$ be an endomorphism. Suppose that, for some $k \geq 0$, the induced chain map $F = (f^k)_* : C_*^A(U) \rightarrow C_*(U)$ has homomorphisms*

$$H_q : C_q^A(U) \longrightarrow C_{q+1}(U) \quad (0 \leq q < n)$$

whose output chains have a common finite bound and which satisfy

$$\partial H_q + H_{q-1} \partial = F_q - c_q \quad (0 \leq q < n),$$

where $H_{-1} = 0$ and the differential on C_0 is zero in this unaugmented equation. Let $h_1, h_2 : U \rightarrow U$ be homomorphisms with elementwise commuting images, and put $h(u) = h_1(u)h_2(u)$. Then there is a finite bound B such that every n -cycle $z \in C_n^A(U)$ satisfies

$$[(hf^k)_* z] = [(h_1 f^k)_* z] + [(h_2 f^k)_* z] \quad \text{in } H_n(C_*^B(U)). \quad (6.1)$$

One bound B may be chosen for any fixed finite collection of pairs (h_1, h_2) satisfying these hypotheses.

Proof. Extend H by zero in all degrees $q \geq n$, and define the chain map

$$F' = F - \partial H - H \partial.$$

Then $F'_0 = c_0$ and $F'_q = 0$ for $1 \leq q < n$. The truncation $H_n = 0$ is important: if z is an n -cycle, it gives

$$F'z = Fz. \quad (6.2)$$

Although the two maps need not agree on arbitrary n -chains, this equality is exactly what will be needed. A common finite output bound controls F, F' and H in every degree used below; taking boundaries does not enlarge a bound.

We use the classical Alexander–Whitney and shuffle comparison in the setting of Eilenberg–Zilber [3]. We spell out the chain maps and their support control: the later product-module argument requires one finite set of simplex types across all coordinates. The Alexander–Whitney diagonal is

$$D(u_0, \dots, u_q) = \sum_{i=0}^q (u_0, \dots, u_i) \otimes (u_i, \dots, u_q). \quad (6.3)$$

We use the tensor differential

$$\partial(a \otimes b) = \partial a \otimes b + (-1)^{\deg a} a \otimes \partial b.$$

In the boundary of (6.3), terms on the two sides of an interior cut cancel, and the remaining terms give $D\partial$. Thus D is a chain map. It is well defined on normalized chains: if the input has consecutive equal vertices, every summand has a degenerate factor. Both tensor factors retain the input bound.

The shuffle map

$$S : C_*(U) \otimes C_*(U) \longrightarrow C_*(U \times U)$$

sends $(u_0, \dots, u_p) \otimes (v_0, \dots, v_q)$ to the signed sum over monotone lattice paths from $(0, 0)$ to (p, q) . A path contributes the tuple obtained by replacing each of its vertices (i, j) by (u_i, v_j) . Its sign

is $(-1)^N$, where N counts pairs of steps in which a vertical step precedes a horizontal step. In its boundary, deleting a corner pairs two paths that traverse the corner in opposite orders; these contributions cancel. The remaining faces are precisely the shuffle of the tensor differential, so S is a chain map. If either input is degenerate, every path has a step with identical endpoint pairs and hence gives a degenerate output. The map therefore descends to normalized chains and preserves the bound in each coordinate.

Write Δ for the chain map of the diagonal homomorphism $U \rightarrow U \times U$. There is a chain homotopy R satisfying

$$\partial R + R\partial = \Delta - SD \quad (6.4)$$

and preserving the bound in both coordinates. Here is a direct construction that also verifies this control. On vertices take $R_0 = 0$, since Δ and SD agree there. Suppose R has been defined in lower degrees, and let $a = (u_0, \dots, u_q)$ be a normalized basis simplex of positive degree, with vertex set V . The chain

$$\Delta a - SDa - R\partial a$$

is a cycle: apply the previously established homotopy equation to ∂a . It lies in the full tuple complex on $V \times V$. That complex is contracted by prepending any fixed vertex, so prepending (u_0, u_0) fills the displayed cycle. Choose this filling as $R(a)$ and extend linearly on the normalized basis. Every coordinate vertex in this construction belongs to V , which proves the claimed bound. This construction takes place on the normalized basis; degenerate faces have already been set to zero.

Apply (6.4) after F . Naturality of D for the vertex map f^k gives a controlled chain homotopy from ΔF to $S(F \otimes F)D$. To replace F by F' , use the degree-one tensor operator K defined on homogeneous tensors by

$$K(a \otimes b) = H(a) \otimes F(b) + (-1)^{\deg a} F'(a) \otimes H(b). \quad (6.5)$$

The tensor differential and the chain-map identities for F, F' give

$$\begin{aligned} \partial K + K\partial &= (F - F') \otimes F + F' \otimes (F - F') \\ &= F \otimes F - F' \otimes F'. \end{aligned}$$

In particular the sign in the second term of (6.5) cannot be dropped. The operator SKD is controlled separately in both coordinates: it uses only the already controlled maps F, F', H and the bound-preserving maps D, S . No equivariance of H enters this calculation. Explicitly, the composite homotopy $L = RF + SKD$ satisfies

$$\partial L + L\partial = \Delta F - S(F' \otimes F')D.$$

Let z be an n -cycle. Since F' vanishes in degrees $1, \dots, n-1$, only the terms of bidegrees $(0, n)$ and $(n, 0)$ survive in $(F' \otimes F')Dz$. The degree-zero factor always maps to (1) . Summing the endpoint terms over the whole cycle first, and then using (6.2), gives

$$\begin{aligned} (F' \otimes F')Dz &= (1) \otimes F'z + F'z \otimes (1) \\ &= (1) \otimes Fz + Fz \otimes (1). \end{aligned}$$

This also covers $n = 1$, when there are no intermediate bidegrees. Shuffling a vertex against a chain simply holds that coordinate fixed.

Finally, commuting images make

$$m : U \times U \longrightarrow U, \quad m(u, v) = h_1(u)h_2(v)$$

a homomorphism. If two pairs have coordinate differences in B_1 and B_2 , their image difference belongs to $h_1(B_1)h_2(B_2)$. Indeed, the factors from the two image subgroups commute, allowing the two differences to be grouped. This finite set is symmetric and contains 1. Thus applying m_* to all the preceding homotopies retains finite control and proves (6.1): the finite chain m_*Lz fills the difference of its left- and right-hand cycle representatives. For a finite collection of pairs, take the union of the finitely many resulting output bounds. $\square$

6.3 One iterate fills every cycle of a fixed bound

We now combine the additivity calculation with the two finite idempotent systems. The exponent and the output bound in the next statement are chosen before the input cycle, which is the uniformity needed when passing later to products of coefficient modules.

Proposition 6.2. *Assume the hypotheses of Theorem 5.1. For every finite bound $A \subseteq U$ and every integer $n \geq 0$, there are an integer $k \geq 0$ and a finite bound $B \subseteq U$ such that, for every reduced integral n -cycle $z \in C_n^A(U)$, there is a finite chain $y \in C_{n+1}^B(U)$ satisfying*

$$\partial y = (f^k)_* z.$$

The chain y may depend on z , with no uniform restriction on its length or coefficients.

Proof. We argue by induction on n . By the finite-presentation factorization $f = ba$ in Theorem 5.1, the subgroup $b(P)$ is finitely generated and contains $f(U)$. Choose a finite symmetric generating set S for $b(P)$, with 1 adjoined. For every $u \in U$, choose a finite edge path from 1 to $f(u)$ whose consecutive differences belong to S . Its chain $H_0(u)$ satisfies

$$\partial H_0(u) = (f(u)) - (1).$$

Extend H_0 linearly. For a reduced zero-cycle z , the constant vertex terms cancel, so $\partial H_0(z) = f_* z$. All these paths use the same bound S , even though their lengths can be arbitrarily large. This proves the assertion for $n = 0$.

Suppose $n \geq 1$ and the assertion holds in all degrees less than n , for every finite input bound. We first construct the partial controlled homotopy in Lemma 6.1 on $C_*^A(U)$. Start with $F = f_*$ and the vertex paths just chosen, which give its homotopy to the constant map in degree zero.

Suppose, for some $1 \leq q < n$, that $F = (f^k)_*$ and controlled operators $H_0, \dots, H_{q-1}$ have already been constructed with

$$\partial H_i + H_{i-1} \partial = F_i - c_i \quad (0 \leq i < q).$$

For each normalized basis q -simplex $a \in C_q^A(U)$, the next homotopy equation asks us to fill

$$F(a) - H_{q-1}(\partial a). \tag{6.6}$$

This is a q -cycle. Its boundary is zero by the preceding homotopy equation applied to ∂a ; the remaining constant-map term $c_{q-1}(\partial a)$ is zero, also when $q = 1$ because the augmentation of a boundary is zero. Moreover, all these cycles have one common finite bound, independent of the basis simplex a : the current F sends A to a finite bound and H_{q-1} is controlled.

Apply the induction assertion in degree q to this common bound. It supplies an integer $r \geq 0$ and another common finite bound such that $(f^r)_*$ sends every cycle (6.6) to a boundary there. Postcompose F and the already chosen H_i by $(f^r)_*$. The existing homotopy equations remain valid, since $f^r(1) = 1$. For each basis simplex choose a finite filling of its newly mapped cycle and use that

filling as $H_q(a)$. Linear extension defines a controlled homomorphism H_q , and the next homotopy equation now holds.

There are only finitely many degrees $q < n$, so this process ends with one power f^k and a controlled homotopy to the constant map through degrees less than n . For $n = 1$ the initial vertex homotopy already suffices. None of these choices is required to be equivariant or computable. In particular, we do not choose one filling on each orbit and then translate it. Uniformity instead comes from the induction assertion applying at each step to *every* cycle in a single bounded complex.

Apply Lemma 6.1 to every orthogonal relation in both $\mathcal{P}_2$ and $\mathcal{P}_3$. There are only finitely many relations, so there is one finite output bound B for all of them; enlarge it if necessary to contain the images of the input bound under every $h_{\ell,E}f^k$. For a fixed n -cycle z , write

$$z_{\ell,E} = [(h_{\ell,E}f^k)_*z] \in H_n(C_*^B(U)).$$

For each $\ell \in \{2, 3\}$, the commuting-image and orthogonal-sum identities, together with Lemma 6.1, give

$$z_{\ell,E+P'} = z_{\ell,E} + z_{\ell,P'}$$

for every orthogonal triple in that finite system. Here the three maps in each relation are evaluated on the common input $f^k(u)$. The two full-identity classes agree because both systems have $h_{\ell,I} = f$:

$$z_I := z_{2,I} = z_{3,I} = [(f^{k+1})_*z].$$

By Lemma 3.1, this single class is annihilated by both 2 and 3. Hence it is zero, since $z_I = 3z_I - 2z_I$. It therefore has a finite filling in $C_*^B(U)$. The exponent $k + 1$ and the bound B were chosen independently of z , proving the required assertion in degree n and completing the induction. $\square$

Proposition 6.2 is a statement about ordinary integral chains with uniform finite simplex types. Its homotopies need not be equivariant, and it asserts no decision procedure for the word problem of U . The next step uses precisely this form of control to assemble coordinatewise fillings with product coefficients.

7 From controlled fillings to finite skeleta

Proposition 6.2 concerns ordinary integral chains. We will apply it coordinate by coordinate to products of regular modules, keeping the finite set of simplex types common to all coordinates. We first explain why the resulting homology vanishing yields geometric, rather than merely homological, finiteness.

All coefficient modules in this section are right modules, and all group homology is integral. For a group J and a set $\mathcal{I}$, write $(\mathbb{Z}J)^{\mathcal{I}}$ for the direct *product* of copies of the right regular module, with coordinatewise action.

7.1 A geometric product-module criterion

The following argument combines the product-module finiteness method of Bieri–Eckmann and Brown [1, 2] with the cellular passage to geometric finiteness. We give the argument in the form needed here, including its finite-presentation hypothesis.

Lemma 7.1. *Let J be a finitely presented group. If*

$$H_n(J, (\mathbb{Z}J)^{\mathcal{I}}) = 0 \quad \text{for every set } \mathcal{I} \text{ and every } n \geq 2,$$

then J is of type F_∞ .

Proof. We construct nested, n -dimensional CW complexes X_n , for $n \geq 2$, with fundamental group J , finite n -skeleton, and $(n - 1)$ -connected universal cover. Each passage from X_n to X_{n+1} will attach finitely many $(n + 1)$ -cells. A finite presentation complex supplies X_2 .

Suppose that X_n has been constructed, and put $Z = \tilde{X}_n$. Thus Z is an n -dimensional, $(n - 1)$ -connected free J -complex. By successively attaching J -orbits of cells of dimensions at least $n + 1$ to kill its homotopy groups, extend Z to a contractible free J -complex Y . There is no finiteness requirement on these auxiliary attachments. Put $R = \mathbb{Z}J$. The augmented cellular chain complex $C_*(Y)$ is a free left R -resolution of $\mathbb{Z}$, and $C_q(Y) = C_q(Z)$ is a finite-rank free R -module for $q \leq n$.

Let $\mathcal{I}$ be the set of all cellular n -cycles of Z , and let $M = R^{\mathcal{I}}$. If C is a finite-rank free left R -module, the coordinate map is an isomorphism

$$M \otimes_R C \longrightarrow C^{\mathcal{I}}, \quad (m_z)_{z \in \mathcal{I}} \otimes c \longmapsto (m_z c)_{z \in \mathcal{I}}. \quad (7.1)$$

Using this identification in degrees n and $n - 1$, the family whose z -coordinate is the cycle z defines a cycle $\xi \in M \otimes_R C_n(Y)$. The hypothesis $H_n(J, M) = 0$ gives

$$\xi = \partial y \quad \text{for some } y \in M \otimes_R C_{n+1}(Y).$$

Choose a free R -basis of $C_{n+1}(Y)$. A tensor is a finite sum, and each chain uses only finitely many basis elements. Consequently we can write

$$y = \sum_{j=1}^s m_j \otimes c_j$$

using finitely many basis elements $c_1, \dots, c_s$, even though $C_{n+1}(Y)$ need not have finite rank. Taking the z -coordinate of the boundary equation gives

$$z = \sum_{j=1}^s (m_j)_z \partial c_j.$$

Thus the module of cellular n -cycles in Z is generated over R by $\partial c_1, \dots, \partial c_s$.

Because Z is $(n - 1)$ -connected and $n \geq 2$, the natural Hurewicz map $\pi_n(Z) \rightarrow H_n(Z; \mathbb{Z})$ is an isomorphism [5, Theorem 4.37 and p. 370]. Naturality with respect to deck transformations makes it an isomorphism of R -modules, using the canonical change of basepoint in the simply connected space Z . Since Z has no $(n + 1)$ -cells, its n th homology is its module of cellular n -cycles. Choose maps of n -spheres representing the displayed finite list of generators, and attach their J -translates to Z . Downstairs this attaches finitely many $(n + 1)$ -cells to X_n ; upstairs the resulting complex is n -connected. It is the universal cover of the resulting complex X_{n+1} , since cells of dimension at least three do not change the fundamental group. This completes the induction.

Let $X = \bigcup_{n \geq 2} X_n$. It has fundamental group J and finitely many cells in every dimension. The universal cover of X is weakly contractible. It is connected and simply connected, and for $q \geq 2$, cellular approximation [5, Theorem 4.8] homotopes any map $S^q \rightarrow \tilde{X}$ into the q -skeleton of $\tilde{X}$, which is already present in $\tilde{X}_q$. That class is killed in $\tilde{X}_{q+1}$. A weakly contractible CW complex is contractible by Whitehead's theorem [5, Theorem 4.5], so X is a $K(J, 1)$. $\square$

7.2 The ascending HNN sequence

For the rest of the section assume the hypotheses of Theorem 5.1, and set $J = T(U, f)$. We will compute the homology in Lemma 7.1 using a tree for J . We give the construction explicitly to fix the direction of the coefficient map.

Form the direct limit

$$L = \varinjlim (U \xrightarrow{f} U \xrightarrow{f} U \xrightarrow{f} \dots),$$

and denote its stage maps by $\iota_i : U \rightarrow L$, for $i \geq 0$. They satisfy $\iota_i = \iota_{i+1}f$. Since f is injective, each ι_i is injective and $L = \bigcup_{i \geq 0} \iota_i(U)$ is an increasing union. The map

$$\sigma(\iota_i(u)) = \iota_{i+1}(u)$$

is an automorphism: its inverse sends $\iota_i(u)$ to $\iota_i(f(u))$. In the semidirect product adjoining a letter t with $txt^{-1} = \sigma(x)$ for $x \in L$, we have

$$t^{-1}\iota_0(u)t = \iota_0(f(u)).$$

This gives a homomorphism from J to that semidirect product. Conversely, the maps $\iota_i(u) \mapsto t^i u t^{-i}$ are compatible with the direct system and with σ , and give the inverse homomorphism. Thus U embeds in J . The same description shows that every element of J can be written

$$t^p u t^{-q}, \quad u \in U, \quad p, q \geq 0. \quad (7.2)$$

Indeed, an element of $L \rtimes \langle t \rangle$ has the form $\iota_p(u)t^m$; increasing p if necessary makes $q = p - m$ nonnegative.

Consider the graph whose vertices and oriented edges are both indexed by the left cosets jU , with the edge indexed by jU running from jU to jtU . The terminal vertex is well defined because

$$ut = tf(u) \quad (u \in U). \quad (7.3)$$

The homomorphism $J \rightarrow \mathbb{Z}$ that kills U and sends t to 1 defines a height on the vertices. Each edge increases height by one, and each vertex has exactly one outgoing edge. If $j = t^p u t^{-q}$ is as in (7.2), the forward path of length q from jU reaches $t^p U$. Every forward ray therefore meets the ray $U, tU, t^2U, \dots$, proving connectedness. An undirected cycle would have a vertex of least height with two outgoing edges, which is impossible. The graph is a tree. Left multiplication gives a J -action with one vertex orbit and one edge orbit, and both the chosen vertex U and the chosen edge from U to tU have stabilizer U .

Let $C_*(U)$ be the normalized homogeneous resolution introduced in the preceding section. Since $\mathbb{Z}J$ is free as a right $\mathbb{Z}U$ -module, induction preserves its exactness. We obtain the free left $\mathbb{Z}J$ -resolution

$$P_* = \mathbb{Z}J \otimes_{\mathbb{Z}U} C_*(U)$$

of the permutation module $\mathbb{Z}[J/U]$. The two endpoint maps on this resolution are the identity and the map

$$E(j \otimes c) = jt \otimes f_*(c).$$

The second is well defined: for $u \in U$,

$$jut \otimes f_*(c) = jtf(u) \otimes f_*(c) = jt \otimes f_*(uc).$$

It is a left $\mathbb{Z}J$ -linear chain map, lifting the terminal-vertex map $jU \mapsto jtU$.

The mapping cone of $1 - E : P_* \rightarrow P_*$, with its natural augmentation, is a free left $\mathbb{Z}J$ -resolution of $\mathbb{Z}$. To see exactness, first take the homology of P_* : it is concentrated in degree zero and equal to $\mathbb{Z}[J/U]$. The remaining two-term complex is the augmented cellular chain complex of the tree, which is exact. Equivalently, the long exact sequence of the cone identifies its homology with that of this two-term complex.

For an arbitrary right $\mathbb{Z}J$ -module M , tensoring this free resolution with M yields the long exact sequence

$$\begin{aligned} H_n(U, M) &\xrightarrow{1-F} H_n(U, M) \longrightarrow H_n(J, M) \\ &\longrightarrow H_{n-1}(U, M) \xrightarrow{1-F} H_{n-1}(U, M), \end{aligned} \quad (7.4)$$

where M is restricted to U in the outer terms. On the chain complex $M \otimes_{\mathbb{Z}U} C_*(U)$, the map F is

$$F(m \otimes c) = mt \otimes f_*(c). \quad (7.5)$$

In particular the coefficient multiplier is t , not t^{-1} .

7.3 Products of regular modules

Fix any set $\mathcal{I}$ and take $M = (\mathbb{Z}J)^{\mathcal{I}}$. We claim that F induces a locally nilpotent endomorphism of $H_i(U, M)$ for every $i > 0$: each individual homology class is killed by some power of F .

Recall that a bound is a finite symmetric subset $A \subseteq U$ containing 1, and $C_q^A(U)$ is generated by the normalized tuples $(u_0, \dots, u_q)$ with $u_i^{-1}u_j \in A$ for all i, j . Let $\mathcal{S}_q(A)$ be their finite set of types up to left translation. The first vertex can be translated to 1, after which all other vertices lie in A , proving finiteness. As $C_q^A(U)$ is free over $\mathbb{Z}U$ on these types, there are natural identifications

$$\begin{aligned} M \otimes_{\mathbb{Z}U} C_q^A(U) &\cong \bigoplus_{s \in \mathcal{S}_q(A)} M \\ &\cong \prod_{\alpha \in \mathcal{I}} \left(\bigoplus_{s \in \mathcal{S}_q(A)} \mathbb{Z}J \right). \end{aligned} \quad (7.6)$$

The second equality interchanges a product with a *finite* direct sum; no analogous interchange with an infinite direct sum is being used.

Here is a concrete interpretation of (7.6). The free abelian basis of $\mathbb{Z}J \otimes_{\mathbb{Z}U} C_q^A(U)$ consists of normalized tuples with $u_0, \dots, u_q \in U$ of the form

$$(ju_0, \dots, ju_q), \quad j \in J, \quad u_a^{-1}u_b \in A \text{ for all } a, b.$$

Each tuple lies in one left coset of U . Expanding the group-ring coefficients, a tensor chain therefore gives, in each coordinate α , a finite ordinary integral chain in such coset tuples. There is one bound A for every coordinate. Conversely, any family of coordinatewise finite chains with a common bound defines a tensor chain by (7.6). Their lengths, coefficients, and group-ring supports need not admit any bound independent of α .

Every chain in $M \otimes_{\mathbb{Z}U} C_q(U)$ uses finitely many simplex types and consequently belongs to a bounded subcomplex of this form. Indeed, take the union of the finitely many relative vertex differences in its tensor factors, then add their inverses and 1. The inclusion of this bounded complex is injective degree by degree, since its types are a subset of the free basis of $C_q(U)$.

Now let $z \in M \otimes_{\mathbb{Z}U} C_i(U)$ be a cycle, where $i > 0$, and choose one input bound A for it. In each coordinate α , split its ordinary chain into its finitely many left-coset components. Each component is a cycle, since the boundary preserves the coset and the coset chain groups form a direct sum. Choose an origin j in each nonzero component's coset. Left translation by j^{-1} identifies the component with an ordinary cycle in $C_i^A(U)$.

Proposition 6.2 supplies an exponent $k \geq 0$ and a finite output bound B that work for *every* such cycle, in every coordinate and every coset. By (7.5), the map F^k changes the origin j to jt^k and applies f^k to the relative vertices. Indeed, on tuples the single-step formula is simply

$$(ju_0, \dots, ju_i) \mapsto (jtf(u_0), \dots, jtf(u_i)) = (ju_0t, \dots, ju_it).$$

This also verifies that the formula is independent of the chosen origin.

For each component, choose an ordinary finite filling of its relative image under f^k in $C_{i+1}^B(U)$, and translate this filling by jt^k . Sum the fillings within each coordinate. The result is still finite in that coordinate because there were only finitely many nonzero coset components. Different input cosets may have the same output coset; in that case their fillings are simply added. Across all coordinates the bound is still the single finite set B . Equation (7.6) therefore assembles these fillings into a chain

$$y \in M \otimes_{\mathbb{Z}U} C_{i+1}^B(U).$$

The boundary and $F^k z$ lie in the bounded complex for $D = B \cup f^k(A)$, a finite symmetric set containing 1. They agree in every coordinate by construction. The coordinate identification (7.6) for D is injective, so $\partial y = F^k z$ in the tensor complex. Thus $F^k[z] = 0$, proving the claimed local nilpotence. This argument uses ordinary fillings chosen separately in each coset; it requires no equivariance of the homotopies or fillings in Proposition 6.2.

7.4 Completion of the torus theorem

If an endomorphism F of an abelian group is locally nilpotent, then $1 - F$ is invertible. On an element x , its inverse is

$$x + Fx + F^2x + \dots,$$

where only finitely many terms are nonzero. The sum defines a homomorphism, and telescoping verifies both inverse identities. It follows that the two maps $1 - F$ adjacent to $H_n(J, M)$ in (7.4) are isomorphisms for $n \geq 2$. Exactness gives

$$H_n(J, (\mathbb{Z}J)^{\mathcal{I}}) = 0 \quad (n \geq 2).$$

Only positive-degree local nilpotence was needed: no assertion about $H_0(U, M)$ enters this conclusion.

By Lemma 5.2, J is finitely presented. The set $\mathcal{I}$ was arbitrary, so Lemma 7.1 proves that J is of type F_∞ . The direct-limit construction already proved that U embeds in J . This completes the proof of Theorem 5.1.

8 Proof of the embedding theorem

We now combine the finite-presentation construction with the geometric criterion.

Proof of Theorem 1.1. Let U be the universal finitely presented group supplied by Lemma 2.2. Corollary 4.2 gives an injective endomorphism $f : U \rightarrow U$ and, for each $\ell \in \{2, 3\}$, homomorphisms $h_{\ell, E} : U \rightarrow U$ indexed by $E \in \mathcal{P}_\ell$. Their full-identity maps are both f , and they satisfy all the elementwise commutation and orthogonal-sum identities in Theorem 5.1.

The factorization hypothesis of that theorem is automatic here: take $P = U$, $a = \text{id}_U$, and $b = f$. Hence $H = T(U, f)$ has type F_∞ . Since f is injective, the natural map $U \rightarrow H$ is injective, as established in Section 7. For every finitely presented group G , choose an embedding $G \rightarrow U$. Its composite with $U \rightarrow H$ is the required embedding. The construction of H is independent of this last choice and of G . $\square$

The construction isolates a useful distinction: finite presentation is needed for the universal group and its diagram envelope, while the faithful models in tensor product algebras are used only to prove injectivity. The higher-dimensional finiteness is obtained afterwards, from the two finite systems and one ascending HNN extension.

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